

# **The Anachronistic Testing-Date Bias of Machine Learning Strategies**

**Federico Tiberga**

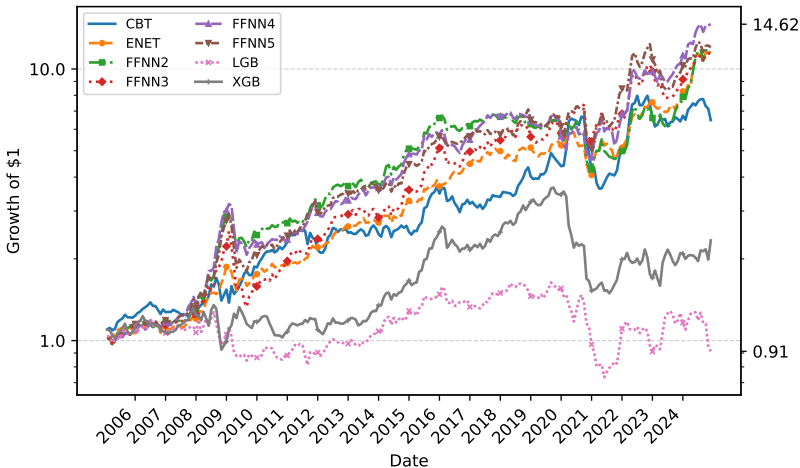
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# Motivation: Machine Learning in Asset Pricing

Cumulative Gross Performance of Long-Short Strategies



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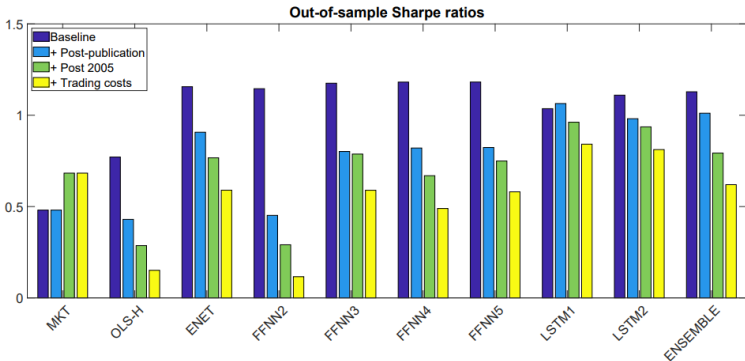
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- ▶ **But how much of this profitability is achievable in practice?**

# Motivation: ML Performance Under Realistic Scenarios

ML performance decomposition Azevedo, Hoegner, and Velikov (2023)



- ▶ The Sharpe ratio of all ML strategies decreases when accounting for realistic scenarios, and only complex models over-perform the market.

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- ▶ **Yet existing backtests** treat the predictive technology as constant and available throughout the sample, even in periods **predating the actual invention or large-scale feasibility** of the underlying algorithms.
- ▶ **This creates a gap**: anachronistic testing-date bias may further inflate reported performance **beyond what transaction costs and post-publication decay alone can explain**.

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- ▶ **The concern:** if markets had no opportunity to learn about and arbitrage the strategy, **full-sample performance can overstate the returns** an investor could have earned after public release.
- ▶ **This paper asks** whether part of the documented outperformance reflects **anachronistic testing-date bias** rather than persistent economic alpha.

# What Is Anachronistic Testing-Date Bias?

- ▶ **Definition:** anachronistic testing-date bias occurs when the information set  $\mathcal{I}_t$  used at time  $t$  includes data transformations, predictors, or model architectures that were **not economically or technologically feasible at time  $t$** .

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  - ▶ **Market awareness:** a method exists, but markets have not yet had time to absorb it.
- ▶ **In this paper, we study model feasibility:** we assign each model its public release date and compare its performance before and after that date.

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  - ▶ Pooled return-shift regression: post-release decline of 0.512% per month (significant at the 5% level).
  - ▶ Pooled factor-adjusted regression: post-release alpha decline of 0.622% per month (significant at the 10% level).

# Machine learning models

# Models Considered

| Model | Description                        | Public release |
|-------|------------------------------------|----------------|
| ENET  | Elastic Net                        | Mar 2005       |
| FFNN  | Feedforward Neural Networks (Adam) | Dec 2014       |
| XGB   | XGBoost                            | Mar 2016       |
| LGB   | LightGBM                           | Dec 2017       |
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- ▶ The empirical interpretation is therefore an average post-release change in performance, rather than an immediate, discrete jump.

# Data and Methodologies

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- ▶ **Trading frictions:** Net returns account for trading costs, estimated using the low-frequency bid-ask spread estimator of Chen and Velikov (2023).

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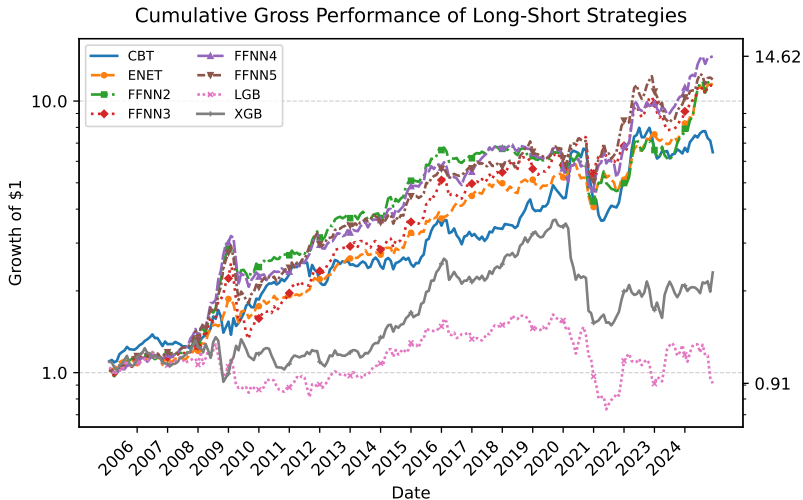
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- ▶ **Third specification:** a **pooled post-release alpha-shift regression** constrains the post-release alpha shift to be common across models, increasing statistical power at the cost of flexibility.
- ▶ All three specifications are estimated directly on net long-short returns, **avoiding the inferential complications of two-step procedures** Pagan (1984).

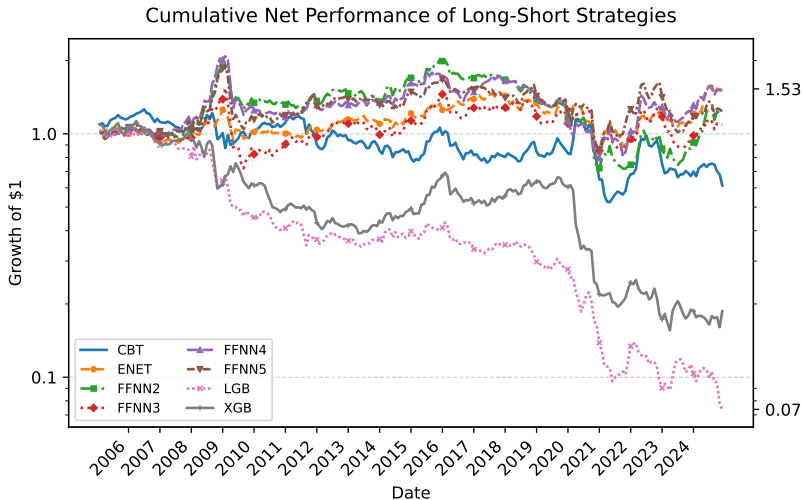
# Results

# Gross Performance of Long-Short Portfolios



- Most models yield positive cumulative gross returns, with feedforward neural networks, elastic net, and CatBoost being statistically significant at the 1% level.

# Net Performance of Long-Short Portfolios



- ▶ Incorporating transaction costs leads to a significant reduction in returns. None of the strategies produces a statistically significant positive net return.

# Gross and Net Performance of Long-Short Portfolios

| Model | Gross    |        |      | Net       |         |       |
|-------|----------|--------|------|-----------|---------|-------|
|       | Ret%     | [t]    | SR   | Ret%      | [t]     | SR    |
| FFNN4 | 1.263*** | [3.75] | 0.84 | 0.307     | [0.92]  | 0.21  |
| FFNN5 | 1.179*** | [3.54] | 0.79 | 0.222     | [0.67]  | 0.15  |
| FFNN2 | 1.163*** | [3.66] | 0.82 | 0.211     | [0.67]  | 0.15  |
| FFNN3 | 1.153*** | [3.50] | 0.79 | 0.169     | [0.51]  | 0.11  |
| ENET  | 1.102*** | [4.46] | 1.00 | 0.249     | [1.01]  | 0.23  |
| CBT   | 0.919*** | [2.72] | 0.61 | -0.066    | [-0.20] | -0.04 |
| XGB   | 0.504    | [1.45] | 0.32 | -0.541    | [-1.52] | -0.34 |
| LGB   | 0.089    | [0.27] | 0.06 | -0.951*** | [-2.87] | -0.64 |

- The decrease in net returns relative to gross returns is substantial and consistent across all models. The FFNN architectures and ENET remain slightly positive, but statistically insignificant.

# Pooled Post-Release Return-Shift Regression

- ▶ In the spirit of McLean and Pontiff (2016), we pool net long-short returns across all models and estimate a single common return shift after public release:

$$r_{i,t} = \alpha_i + \beta POST_{i,t} + \varepsilon_{i,t},$$

where  $POST_{i,t} = 1$  if month  $t$  occurs after the public release of model  $i$ , and  $\beta$  is the pooled average post-release return shift, constrained to be common across models.

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- ▶ **Interpretation:**  $\hat{\beta}$  captures the average post-release shift in net returns, without factor controls. Standard errors use Newey–West HAC estimators.

# Pooled Post-Release Return-Shift Regression: Results

| $\hat{\beta}\%$ | [t]     | p-value | N    | $R^2$ |
|-----------------|---------|---------|------|-------|
| -0.512**        | [-2.05] | 0.040   | 1912 | 0.009 |

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- ▶ This specification is the **simplest and most transparent test** of our central hypothesis: it does not adjust for risk and applies a **uniform post-release shift across all models**.
- ▶ The negative and statistically significant pooled estimate provides the **clearest evidence in this paper** that machine-learning return premiums decline after public release.

# Model-Specific Factor-Adjusted Alpha-Shift Regression

- ▶ To isolate post-release alpha decay within each model and prevent shifts in systematic risk exposures from contaminating the estimates, we estimate separately for each model  $i$ :

$$r_{i,t} = \alpha_i + \Delta\alpha_i POST_{i,t} + \beta_i' F_t + \Delta\beta_i' (POST_{i,t} \times F_t) + \varepsilon_{i,t},$$

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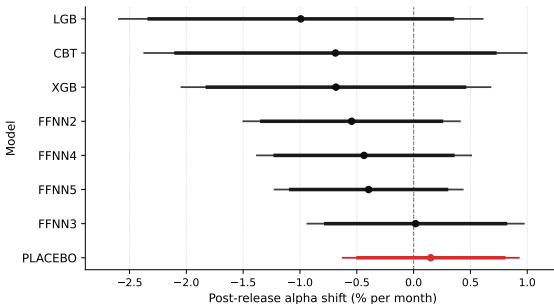
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- ▶ The interaction terms  $\Delta\beta_i' (POST_{i,t} \times F_t)$  allow factor loadings to shift at release, ensuring that  $\Delta\hat{\alpha}_i$  is not contaminated by changes in systematic risk exposures.

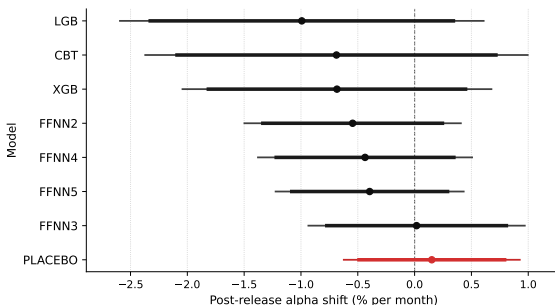
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- **Placebo check:** assigning a synthetic March 2015 release date to ENET yields +0.151% ( $p = 0.705$ ). The placebo result does not replicate the negative pattern, reducing concern about a generic post-2015 time trend.

# Pooled Factor-Adjusted Alpha-Shift Regression

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- ▶ The restricted specification sets  $\Delta\beta_i = 0$ , holding factor loadings constant after release. This trades flexibility for greater statistical power.

# Pooled Factor-Adjusted Alpha-Shift Regression: Results

| Specification                    | $\Delta\hat{\alpha}\%$ | [t]     | p-value | $R^2$ |
|----------------------------------|------------------------|---------|---------|-------|
| Full (allows $\Delta\beta$ )     | -0.523                 | [-1.53] | 0.126   | 0.301 |
| Restricted ( $\Delta\beta = 0$ ) | -0.622*                | [-1.86] | 0.063   | 0.257 |

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- ▶ The close proximity of the full and restricted estimates further indicates that changes in factor loadings are not the main driver of the post-release result.
- ▶ Collectively, the three specifications point to lower post-release performance, with the strongest statistical support from the pooled return-shift regression and weaker but directionally consistent evidence in the factor-adjusted specifications.

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- ③ The strongest evidence of post-release decay comes from the pooled return-shift regression:
  - ▶ Average post-release decline of 0.512% per month ( $p = 0.040$ ).
  - ▶ Pooled factor-adjusted regressions:  $-0.523\%$  and  $-0.622\%$  per month, directionally consistent.
  - ▶ Individual model estimates: six of seven negative, but not individually significant.

## Conclusion (continued)

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- ⑥ **Future work** could replace fixed release dates with a **continuous diffusion index** based on academic publications, software releases, and computational-feasibility measures, modelling post-release decay as a **gradual process** rather than a sudden break.

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